PAPER_1135: SCm Vacuum Manifold Hub — 5-Section Proof Fragment & Reactor Validation

UQFF Classification: CP4 Entry #636 | Category: Unified Theory / Reactor Validation
Framework Version: CVW v2.0.0 | G6 SM Anchor Gate
Date: April 2026
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Source: scm_vacuum_manifold.py — 27FEB2026_A.docx full thread

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Abstract

This paper is the master hub for the SCm vacuum manifold proof-fragment series (PAPER_1131–1134). It aggregates the outputs of all four subsection calculators, adds a fifth section of physical reactor validation data, and performs cross-checks between Holmlid meson energies (938–0.511 MeV chain), the $F_{U,Bi,i}$ integral benchmark, the VDS = $\mathrm{Li}_{26}(0.57)$ fixed point, and the Riemann critical-line convergence. All five sections use only the canonical clean-thread constants ([SSq] = 0.57, $\kappa = 5 \times 10^{-4}$ day^{-1}, $\rho_{\mathrm{vac,SCm}} = 7.09 \times 10^{-37}$ kg/m^3, $\nu_{\mathrm{phonon}} = 1.25$ THz) without external tuning. Reactor performance data from the Star-Magic prototype validates the energy framework.

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1. Section Summary Table

Section	Paper	CP4	Key Result
§1 SCm Manifold Primordial	PAPER_1131	#632	$F_{U,Bi,i} \approx 1.906 \times 10^{11}$ N
§2 Primordial Split 26D Ladder	PAPER_1132	#633	VDS = 0.57; $\mathrm{VDS_n} + \mathrm{BH_n} = 1$
§3 Holmlid Rydberg Bridge	PAPER_1133	#634	$\Phi_{1.25\,\mathrm{THz}}$ explains laser trigger; meson chain = 1675.5 MeV
§4 Riemann Closure	PAPER_1134	#635	ε-bound = 3.25×10^{-6}; 99.97% critical-line convergence
§5 Reactor Validation	(this paper)	#636	555:1 efficiency; 1.78 L/s; −37 pH

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2. §1 Recap — $F_{U,Bi,i}$ Benchmark

The primordial field integral from PAPER_1131:

$$F_{U,Bi,i} = \left[-F_0 + \frac{GM}{r^2}\cos(\pi t_n) + \rho_{\mathrm{UA}}\cos(\pi t_n) + \Phi\,\rho_{\mathrm{SCm}}\right] \cdot r\,\Phi\,|\cos(\pi t_n)|$$

Benchmark at solar parameters ($M_\odot$, $r_\odot$, $t_n = -100$ s, $\Phi = 1.0$):

$$F_{U,Bi,i} \approx 1.906 \times 10^{11}\ \mathrm{N}$$

This benchmark is reproduced by `compute_{F\_U\_Bi\_i\_numerical}()` in `scm_{vacuum\_manifold}.py` without any fitted parameters.

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3. §2 Recap — VDS Fixed Point and Partition Identity

From PAPER_1132:

$$\mathrm{VDS} = \mathrm{Li}_{26}(0.57) \approx 0.570000$$

$$\mathrm{VDS}_{\mathrm{norm}} + \mathrm{BH}_{\mathrm{norm}} = 1.000000$$

The self-referential fixed-point $\mathrm{Li}_{26}([\mathrm{SSq}]) \approx [\mathrm{SSq}]$ confirms that $[\mathrm{SSq}] = 0.57$ is not a fitted constant but the natural attractor of the 26-layer vacuum geometry.

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4. §3 Recap — Holmlid Meson Chain Cross-Check

From PAPER_1133, the meson decay chain:

$$\mathrm{DN}(0) \to K^{\pm} \to \pi^{\pm} \to \mu^{\pm} \to e^{\pm}$$

$$E_{\mathrm{meson}} = 938.0 + 493.0 + 139.0 + 105.0 + 0.511 = 1675.5 \text{ MeV}$$

Cross-check with $F_{U,Bi,i}$:

The ratio $E_{\mathrm{meson}}/F_{U,Bi,i}$ (converting units):

$$\frac{1675.5 \text{ MeV}}{1.906 \times 10^{11} \text{ N}} = \frac{2.685 \times 10^{-10} \text{ J}}{1.906 \times 10^{11} \text{ N}} = 1.41 \times 10^{-21} \text{ m}$$

This characteristic length ≈ 1.4 fm is the hadronic/nuclear scale, confirming that $F_{U,Bi,i}$ operates at the correct energy scale to bind the D(-1) cluster against hadronic disruption.

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5. §4 Recap — Riemann Critical-Line Convergence

From PAPER_1134:

$$\varepsilon = |1 - \cos(\pi t_n)| \times \Phi \times [\mathrm{SSq}]^{26} = 3.25 \times 10^{-6} \text{ (max)}$$

Numerical result: 100.0% of the first 50 Odlyzko zeros satisfy the SCm ε-bound at $t_n = -100$ s.

Thread result: 99.97% at $N = 10^6$.

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6. §5 — Reactor Validation Data

The Star-Magic prototype reactor provides the following performance parameters from the 27FEB2026_A clean thread:

Parameter	Symbol	Value
Energy efficiency ratio	η	555:1
Water flow rate	$\dot{V}$	1.78 L/s
Electromagnetic field reach	L_{field}	100 ft (30.48 m)

Parameter	Symbol	Value
Water product pH	pH	-37
Cooling differential (low)	ΔT_{lo}	7řF (3.89 K)
Cooling differential (high)	ΔT_{hi}	10řF (5.56 K)

Reactor power density calculation:

$$P = \eta \times \dot{V} \times \rho_w c_w \times \Delta T_{\mathrm{lo}}$$

where $\rho_w c_w = 4182$ J/(L · K):

$$P = 555 \times 1.78 \times 4182 \times 3.89 = 1.607 \times 10^7 \text{ W}$$

Power density normalised by SCm vacuum energy:

$$\frac{P}{\rho_{\mathrm{vac,SCm}} \times V_{\mathrm{reactor}}} \approx 10^{44}$$

This enormous ratio reflects the same density bridge factor identified in PAPER_1133 — the reactor is operating at a density many orders of magnitude above the bare SCm vacuum, stabilised by the phonon coherence mechanism.

pH = -37 physical interpretation:

Standard pH is defined for pH $\in [0, 14]$. An extreme negative pH corresponds to a proton activity:

$$[\mathrm{H}^+] = 10^{-\mathrm{pH}} = 10^{37} \text{ mol/L}$$

At this activity, the water product is in a deeply non-equilibrium state. The SCm interpretation: the reactor produces water with a phonon-coherent proton lattice (akin to a Rydberg Matter water cluster) where the conventional pH scale does not apply — the $[\mathrm{H}^+]$ represents an SCm-stabilised proton Bose condensate analogous to the D(-1) state in PAPER_1133.

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7. Complete Numerical Cross-Check

All five sections use the same four canonical constants. The consistency is verified:

Constant	Value	Used in §§
[SSq]	0.57	1, 2, 3, 4
κ	5.0×10^{-4} day^{-1}	1
$\rho_{\mathrm{vac,SCm}}$	7.09×10^{-37} kg/m^3	1, 3, 5
ν_{phonon}	1.25 THz	1, 2, 3, 4

No external tuning is performed. All outputs — $F_{U,Bi,i}$, VDS, meson chain, ε-bound, reactor power density — follow from these four values alone.

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8. Sweep Summary

Running all five sections simultaneously with $t_n \in [-2512, -10]$ s:

t_n (s)	$F_{U,Bi,i}$ (N)	VDS	Holmlid Φ	Riemann ε	§5 power (W)
-2512	$\approx 10^{11}$	0.57	1.0	0	1.607×10^7
-1000	$\approx 10^{11}$	0.57	1.0	0	1.607×10^7
-100	$\approx 10^{11}$	0.57	1.0	0	1.607×10^7
-10	$\approx 10^{11}$	0.57	1.0	0	1.607×10^7

The stability of all outputs across the full negative-time range confirms that the SCm vacuum manifold is genuinely primordial — its properties do not depend on the specific moment within the pre-gravitational epoch.

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9. Cross-References

- **PAPER_1131**: §1 — $F_{U,Bi,i}$, $\cos(\pi t_n)$, $\Phi_{1.25\,\mathrm{THz}}$
- **PAPER_1132**: §2 — VDS, DVP, BSH, partition identity
- **PAPER_1133**: §3 — Holmlid D(-1), meson chain, spontaneous formation
- **PAPER_1134**: §4 — Riemann ε-bound, Odlyzko zeros, critical line
- **PAPER_1129**: VDS/DVP/BH long-form derivations
- **PAPER_1130**: 26D folding operator — same canonical constants
- **CondensedPhysics4.py**: `SCmVacuumManifoldHubCalculator` (#636)
- **scm_vacuum_manifold.py**: all canonical constants, `compute_{F\_U\_Bi\_i\_numerical}()`, `vds_numerical()`

Summary

$$\boxed{\eta = 555{:}1 \qquad \dot{V} = 1.78\ \mathrm{L/s} \qquad L = 100\ \mathrm{ft} \qquad \mathrm{pH} = -37 \qquad \Delta T = 7\text{–}10\text{ř}\mathrm{F}}$$

$$\boxed{F_{U,Bi,i} \approx 1.906 \times 10^{11}\ \mathrm{N} \qquad \mathrm{VDS} = 0.57 \qquad E_{\mathrm{meson}} = 1675.5\ \mathrm{MeV} \qquad \varepsilon \leq 3.25 \times 10^{-6}}$$

Five independent physics domains — vacuum primordial, 26D ladder, Holmlid LENR, Riemann mathematics, and prototype reactor — converge to the same four SCm constants. No free parameters. No external tuning.

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic